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Aim:

To implement a Bus Ticket Reservation System using Python classes and constructors.

Problem Statement:

A transport company wants to build a simple bus reservation system.

The system should accept passenger name, destination, and number of seats required.

If seats are available, confirm booking and reduce seat count.

Otherwise, display that seats are full.

The program should repeatedly allow booking and checking available seats using loops.

After completion, display total passengers booked and remaining seats.

Class Definition

class Bus:

Constructor to initialize total seats, available seats, and passenger count

```
def __init__(self, total_seats):  
    self.total_seats = total_seats  
    self.available_seats = total_seats  
    self.total_passengers = 0
```

Method to reserve seats for passengers

```
def reserve_seat(self, name, destination, seats_required):  
    if seats_required <= self.available_seats:  
        self.available_seats -= seats_required  
        self.total_passengers += seats_required  
        print("\nBooking Confirmed")  
        print("Passenger Name:", name)  
        print("Destination:", destination)  
        print("Seats Booked:", seats_required)  
        print("Remaining Seats:", self.available_seats)  
    else:  
        print("\nSeats are full or not enough seats available")
```

```
# Method to display available seats
```

```
def show_available_seats(self):
```

```
    print("Available Seats:", self.available_seats)
```

```
# Creating Bus object with 20 seats
```

```
bus = Bus(20)
```

```
# Menu-driven loop for repeated booking and checking seats
```

```
while True:
```

```
    print("\n==== Bus Reservation System =====")
```

```
    print("1. Book Ticket")
```

```
    print("2. Check Available Seats")
```

```
    print("3. Exit")
```

```
    choice = input("Enter your choice: ")
```

```
    if not choice.isdigit():
```

```
        print("Enter a valid number")
```

```
        continue
```

```
    choice = int(choice)
```

```
if choice == 1:
```

```
    name = input("Enter Passenger Name: ")
```

```
    destination = input("Enter Destination: ")
```

```
    seats = input("Enter number of seats required: ")
```

```
    if not seats.isdigit():
```

```
        print("Seats must be a number")
```

```
        continue
```

```
    bus.reserve_seat(name, destination, int(seats))
```

```
elif choice == 2:
```

```
    bus.show_available_seats()
```

```
elif choice == 3:
```

```
    break
```

```
else:
```

```
    print("Invalid choice")
```

Final Output after completion

```
print("\nTotal Passengers Booked:", bus.total_passengers)
```

```
print("Remaining Seats:", bus.available_seats)
```

Conclusion:

Thus, the Bus Ticket Reservation System is successfully implemented

using Python classes, constructor, loops, and conditional statements.

===== Bus Reservation System =====

1. Book Ticket
2. Check Available Seats
3. Exit

Enter your choice: 1

Enter Passenger Name: samanvith

Enter Destination: delhi

Enter number of seats required: 3

Booking Confirmed

Passenger Name: samanvith

Destination: delhi

Seats Booked: 3

Remaining Seats: 17

Remaining Seats: 17

===== Bus Reservation System =====

1. Book Ticket
2. Check Available Seats
3. Exit

Enter your choice: 2

Available Seats: 17

===== Bus Reservation System =====

1. Book Ticket
2. Check Available Seats
3. Exit

Enter your choice: 3

Total Passengers Booked: 3

Remaining Seats: 17